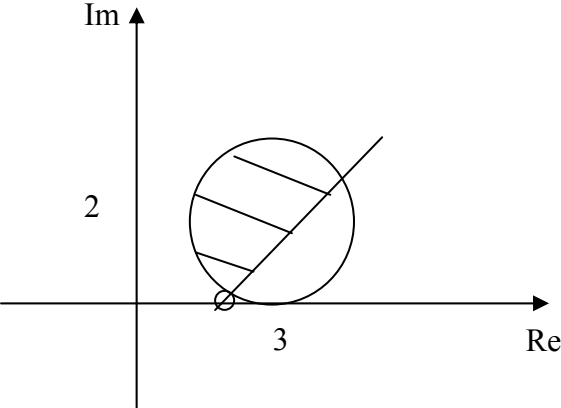


1	$\arg\left(\frac{z-3}{1+i}\right) \geq 0 \Rightarrow \arg(z-3) \geq \arg(1+i)$ a) $\arg(z-3) \geq \frac{\pi}{4}$ $z-3-2i \leq 2$  Least $z-5 = 3 \sin \frac{\pi}{4} = \frac{3}{\sqrt{2}}$ max $z-5 = \sqrt{8} + 2$ (b) angle $SOT = \arg(1+2i) - \arg(4-2i)$ $= \arg\left(\frac{1+2i}{4-2i}\right)$ $= \arg\left(\frac{1}{2}i\right) = \frac{\pi}{2}$	
2 (i)	Net rate of increase of fuel = rate of increase due to fuel pumped in - rate at which fuel is used up $\frac{dx}{dt} = 2\left(\frac{3}{2} - x\right) - kx^2, \text{ where } k > 0$ When $x=1$, $\frac{dx}{dt} = 0$, $k=1$. Hence $\frac{dx}{dt} = 3 - 2x - x^2$.	

(ii)

$$\begin{aligned}\frac{dx}{dt} &= -(x^2 + 2x - 3) \\ \frac{1}{x^2 + 2x - 3} \frac{dx}{dt} &= -1 \\ \int \frac{1}{x^2 + 2x - 3} dx &= -\int dt \\ \int \frac{1}{(x+1) - 2^2} dx &= -\int dt \\ \frac{1}{4} \ln \left| \frac{x-1}{x+3} \right| &= -t + C \\ \left| \frac{x-1}{x+3} \right| &= e^{-4t+4C} \\ \frac{x-1}{x+3} &= Ae^{-4t} \quad \text{where } A = \pm e^{4C}\end{aligned}$$

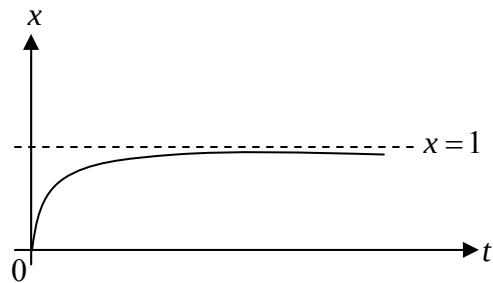
When $t = 0$, $x = 0$, $A = -\frac{1}{3}$,

$$\begin{aligned}-3(x-1) &= (x+3)e^{-4t} \\ x(3+e^{-4t}) &= 3-3e^{-4t} \\ x &= \frac{3(1-e^{-4t})}{3+e^{-4t}}\end{aligned}$$

Alternatively:

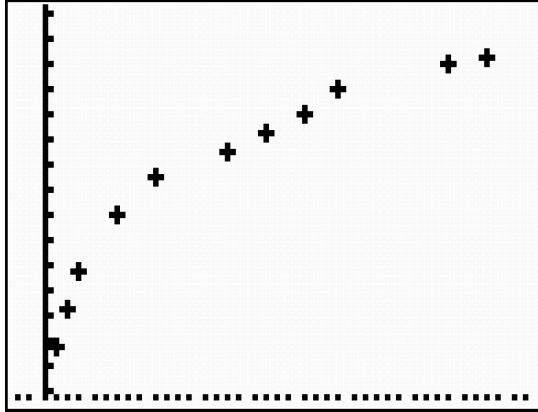
$$\begin{aligned}1 - \frac{4}{x+3} &= -\frac{1}{3}e^{-4t} \\ x &= \frac{12}{3+e^{-4t}} - 3\end{aligned}$$

(iii)



3	$y = a(t + \sin t) \Rightarrow \frac{dy}{dt} = a(1 + \cos t)$ When $y = 0, t = 0$ When $y = 2\pi a, t = 2\pi$ Volume $= \int_0^{2\pi a} \pi x^2 dy = \pi \int_0^{2\pi} a^2 (1 - \cos t)^2 \times a(1 + \cos t) dt$ $= \pi a^3 \int_0^{2\pi} (1 - \cos t)^2 (1 + \cos t) dt = 9.87a^3 \text{ or } 3.14\pi a^3$	
4	(i) Area $= \int_{-3}^1 y dx = 2 \int_{-3}^1 (1-x) \sqrt{(x+3)} dx$ (ii) $u^2 = x+3 \Rightarrow 2u \frac{du}{dx} = 1$ When $x = -3, u = 0$ When $x = 1, u = 2$ Area $= 2 \int_0^2 (4-u^2) u \times 2u du = 4 \int_0^2 (4u^2 - u^4) du$ $= 4 \left[\frac{4}{3} u^3 - \frac{1}{5} u^5 \right]_0^2 = 17 \frac{1}{15} \text{ units}^2$	
4	Solution: (i) Solve using GC, $l: \mathbf{r} = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -7 \\ 16 \\ 13 \end{pmatrix}, \lambda \in \mathbb{R}$	

	(ii) $p_1: \mathbf{r} \cdot \frac{1}{\sqrt{35}} \begin{pmatrix} 5 \\ 3 \\ -1 \end{pmatrix} = \frac{2}{\sqrt{35}}$ $\begin{pmatrix} 5 \\ 3 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = 6$ $p_4: \mathbf{r} \cdot \begin{pmatrix} 5 \\ 3 \\ -1 \end{pmatrix} = 10$ (iii) $\begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} \alpha \\ -1 \\ 2 \end{pmatrix} = 0$ $\Rightarrow \alpha + 2 + 6 = 0 \Rightarrow \alpha = -8$ (iv) $\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} \alpha \\ -1 \\ 2 \end{pmatrix} = \beta$ $\Rightarrow \alpha + 1 = \beta$ A possible set of values is $\alpha = 2, \beta = 3$ ($\alpha = \frac{10}{7}$ and $\beta = \frac{17}{7}$). (v) If $\alpha = \frac{10}{7}$ and $\beta = \frac{17}{7}$, solving the system of linear equations, the 3 planes intersect at the line $l: \mathbf{r} = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -7 \\ 16 \\ 13 \end{pmatrix}, \lambda \in \mathbb{R}$. If $\alpha = \frac{10}{7}$ and $\beta \neq \frac{17}{7}$, the 3 planes do not intersect at l. There is no solution and the 3 planes form a triangular prism.	
5(a)	The scatter diagram gives a good overview of how the two variables are related.	
(b)	(i) $r = 0.9374$	



The scatter diagram shows the relationship between y and x to be non-linear so it is not suitable to find the linear regression line for y on x .

(ii) The linear product moment coefficient between $\ln y$ and $\ln x$ is 0.998.

As r is very close to 1, we can say that the model $\ln y = a + b \ln x$ is appropriate.

$$a = 3.6999, \quad b = 0.07058$$

6

(i) Let X denotes the height of a girl
 $X \sim N(155, 100)$

$$P(155 - a < X < 155 + a) = 0.4$$

$$P(X < 155 - a) = 0.3$$

$$P(Z < -\frac{a}{10}) = 0.3$$

$$-\frac{a}{10} = -0.5244 \Rightarrow a = 5.24$$

range of girls' height is (149.76, 160.24)

(ii)

$$P(X > 170) = P(Z > \frac{170 - 155}{10}) = 0.066807$$

$$\begin{aligned} \text{required probability} &= (0.066807)^2 (1 - 0.066807)^2 \frac{4!}{2!2!} \\ &= 0.0233 \end{aligned}$$

(iii)

	$X_T \sim N(155n, 100n)$ $P(X_T > 150n) = P\left(Z > \frac{150n - 155n}{\sqrt{100n}}\right)$ $= P\left(Z > -\frac{1}{2}\sqrt{n}\right)$ $= 1 \text{ (since } n \text{ is large)}$	
7	$\bar{x} = \frac{-1500}{150} + 120 = 110$ $s^2 = \frac{1}{149} \left[462000 - \frac{(-1500)^2}{150} \right] = 3000$ - To test $H_0 : \mu = 120$ $H_1 : \mu < 120$ (overstated the average time spent) at 1 % level of significance - Test Statistic: Under H_0, $Z = \frac{\bar{X} - \mu_0}{s / \sqrt{n}} \sim N(0, 1)$ $z_{\text{calc}} = \frac{110 - 120}{\sqrt{\frac{3000}{150}}} = -2.236$ [p-value=0.0253] - Critical Region: Reject H_0 if $z_{\text{calc}} < -2.326$ Conclusion: Since $z_{\text{calc}} > -2.326$, it does not lie in the critical region, we do not reject H_0. There is insufficient evidence, at 1% level, that the teacher has overstated the average time spent on mathematics revision per day by a student.	
(ii)	To reject H_0, $\frac{110 - 120}{\sqrt{\frac{3000}{n}}} < -2.326$ $n > 162.3$ least $n = 163$	
(b) (i)	$H_0 : \mu = 4$ (not more than 4) $H_1 : \mu > 4$ at α % level of significance	

	- Under H_0, $T = \frac{\bar{X} - \mu_0}{s / \sqrt{n}} \sim t(7)$ $t_{cal} = \frac{4.08875 - 4}{0.16617 / \sqrt{8}}.$ $P\text{-value} = 0.08731$ - Critical Region: Reject H_0 if $p\text{-value} < \frac{\alpha}{100}$ To reject $H_0 \Rightarrow 0.08731 < \frac{\alpha}{100}$ $\alpha > 8.731$ largest level of significance is 8.73%	
(ii)	Distribution of iron content in substance Y is assumed to be normal.	B1
8	(i) Let A and B denote the average number of dengue fever per month in City A and B respectively. $A \sim P(2.5)$ $B \sim P(4)$ $P(A > 3) = 1 - P(A \leq 3) = 0.242$ (ii) $A + B \sim P(6.5)$ $P(A < 2 / A + B = 4)$ $= \frac{P(A < 2 \cap A + B = 4)}{P(A + B = 4)}$ $= \frac{P(A = 0)P(B = 4) + P(A = 1)P(B = 3)}{P(A + B = 4)}$ $= \frac{0.016037 + 0.04009}{0.11182} = 0.502$ (iii) Let T denote the average number of dengue fever for both cities in 2 months. $T \sim P(13)$ Since $\lambda > 10$ Approximate to $T \sim N(13, 13)$ $P(T < 15) \rightarrow P(T < 14.5) = 0.329$	

	(iv) Let X denote the average number of months out of 12 with at most 3 dengue cases $X \sim B(12, 0.75758)$ $P(X \geq 8) = 1 - P(X \leq 7) = 0.858$ It is not applicable as the number of dengue fever is seasonal and thus it is not applicable over a long period of time.	
9	(a) (i) Let X denote the no of patients out of 20 who will not recover $X \sim B(20, 0.02)$ $P(X > 2) = 1 - P(X \leq 2) = 0.00707$ (ii) Let Y denote the no of sample out of 50 that has more than half patient who will not recover $\bar{Y} \sim N(0.4, \frac{0.392}{50}), CLT$ $P(\bar{Y} > 0.5) = 0.129$ (b) (i) Let Y denote the no of patients out of 50 who will recover $Y \sim B(50, p)$ $P(Y > 48) = 0.64$ $P(Y \leq 48) = 0.36$ From GC, $p = 0.97481 = 0.975$ (to 3 sig fig) (ii) Let Y denote the no of patients out of 50 who will recover $Y \sim B(50, 0.97481)$ Let Y' denote the no of patients out of 50 who will not recover $Y' \sim B(50, 0.02519)$ since $n > 50, p < 0.1, np < 5$	

	approx $Y' \sim P_o(1.2595)$ So $P(Y < 26) = P(Y' \geq 5)$ $= 1 - P(Y' \leq 4) = 0.00191$	
10(a)	Case 1: 3 identical letter 1 way Case 2: 2 identical letters ${}^3C_1 {}^6C_1 \times \frac{3!}{2!} = 54$ Case 3: All different ${}^7C_3 \times 3! = 210$ Total Number of ways = 265	
(i)	Required probability $= \frac{1}{10} + \frac{9}{10} \cdot \frac{1}{10} + \left(\frac{9}{10}\right)^2 \cdot \frac{1}{10} + \left(\frac{9}{10}\right)^3 \cdot \frac{1}{10} + \left(\frac{9}{10}\right)^4 \cdot \frac{1}{10}$ $= \frac{\frac{1}{10} \left(1 - \left(\frac{9}{10}\right)^5\right)}{1 - \frac{9}{10}} = 1 - \left(\frac{9}{10}\right)^5 = 0.40951$	
(ii)	Required probability = $\frac{\left(\frac{9}{10}\right)^7 \left(\frac{1}{10}\right)}{1 - \left(1 - \left(\frac{9}{10}\right)^5\right)} = 0.081$	
(iii)	Required Probability $= \left[\left(\frac{9}{10}\right) \left(\frac{1}{10}\right) + \left(\frac{9}{10}\right)^3 \left(\frac{1}{10}\right) + \left(\frac{9}{10}\right)^5 \left(\frac{1}{10}\right) + \dots \right]$ $= \frac{\left(\frac{9}{10}\right) \left(\frac{1}{10}\right)}{1 - \left(\frac{9}{10}\right)^2} = \frac{9}{19}$	